

How to create OData service for ABAP CDS Views using Annotation

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Shares everyone in this tutorial we will see on how to create SAP OData service for an ABAP CDS views using annotation **@OData.publish: true**. Lets get started

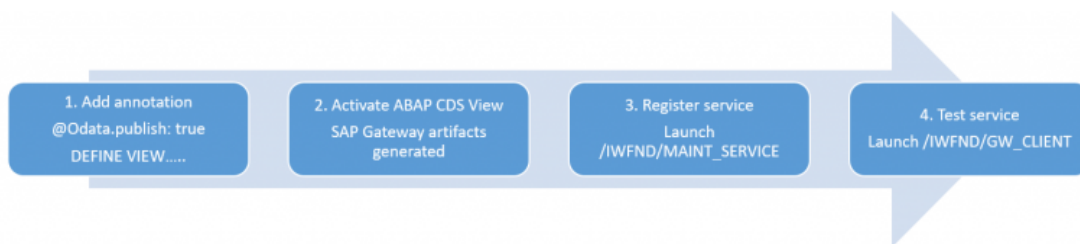
usually create OData services in SAP Gateway system using transaction SAP Gateway Service Builder(SEGW).

ABAP CDS View OData Annotation

introduction of ABAP CDS views in SAP Business Suite 4(S/4 HANA), several thousands of CDS views are added for efficient and faster data access.

and easy way of creating OData services based on CDS views was introduced using the annotation **@OData.publish: true** at view level.

see how we can expose the ABAP CDS View as OData Service using an example and below is the exposure process.



Step-by-Step Procedure

Example #1 – ABAP CDS View with out Parameters

1. Create an ABAP CDS view using ABAP Development Tools in Eclipse. Copy and paste the below code in DDL editor.

```

1. @AbapCatalog.sqlViewName: 'ZV_ODATA_DEMO'
2. @AbapCatalog.compiler.compareFilter: true
3. @AccessControl.authorizationCheck: #NOT_REQUIRED
4. @EndUserText.label: 'OData service on ABAP CDS View'
5.
6.
7. define view ZODATA_DEMO as select from scarr
8. {
9.   key carrid      as AirlineCode,
10.   carrname       as AirlineName,
11.   currcode       as Currency,
12.   url            as AirlineURL
13. }
```

2. Add the annotation `@OData.publish: true` above the DEFINE VIEW statement.

3. After adding the annotation to CDS View save and activate the view.

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```

1. @AbapCatalog.sqlViewName: 'ZV_ODATA_DEMO'
2. @AbapCatalog.compiler.compareFilter: true
3. @AccessControl.authorizationCheck: #NOT_REQUIRED
4. @EndUserText.label: 'OData service on ABAP CDS View'
5.
OData.publish: true

```

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```

define view ZODATA_DEMO as select from scarr

```

```

    by carrid      as AirlineCode,
       carrname    as AirlineName,
       currcode    as Currency,
       url         as AirlineURL
}

```

view should meet following rules for successful OData service generation.

• syntax errors in DDL source code.

• least **one** key element is defined in the SELECT list of the CDS view.

• the name of the CDS view should not exceed 26 characters in length.

After activating the CDS view, following Gateway artifacts will be generated by the SADL framework in back-end.

- Technical service name artifact – `<cds_view_name>_CDS`.
- Gateway Model artifact with name – `<cds_view_name>_CDS`.
- ABAP class with name – `CL_<cds_view_name>`.

You can find all these artifacts in the transaction code – `"/IWBEP/REG_SERVICE"`.

The screenshot shows the 'Display Service' transaction in SAP. It has two tabs: 'Cleanup Cache' and 'Configuration'. The 'Configuration' tab is active, showing the following information:

Service Information	
Technical Service Name	ZODATA_DEMO_CDS
Service Version	1
Description	OData service on ABAP CDS View
External Service Name	ZODATA_DEMO_CDS
Namespace	
Data Provider Class	CL_SADL_GTK_EXPOSURE_DPC
Created By	
Changed By	
Package	SADL_GW_RT_DPC

Extension for Service	
Technical Service Name	
Service Version	0

Model Information	
Technical Model Name	ZODATA_DEMO_CDS
Model Version	1
Description	OData service on ABAP CDS View
Model Provider Class	ZCL_ZODATA_DEMO
Created By	
Changed By	
Package	\$TMP

At the bottom, there are four buttons: 'Create Model', 'Unassign Model', 'Assign Model', and 'Annotation Model'.

5. Launch the transaction code `"/IWFND/MAINT_SERVICE"` to activate the OData service in hub system. Enter **System Alias**, **Technical Service Name** and click the **Get Services** button. The service will be displayed for selection.

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Add Selected Services

Get Services

Filter

System Alias: LOCAL

Technical Service Name: ZODATA_DEMO_CDS

External Service Name:

Version:

External Mapping ID:

Add Selected Services

Select Backend Services

Type	Technical Service Name	Version	Service Description	External Service Name
BEP	ZODATA_DEMO_CDS	1	OData service on ABAP CDS View	ZODATA_DEMO_CDS

the **Add Service** dialog box enter the package name. Technical service name and Technical model name will be suggested. Click on OK to continue.

Add Service

Service

Technical Service Name: ZODATA_DEMO_CDS

Service Version: 1

Description: OData service on ABAP CDS View

External Service Name: ZODATA_DEMO_CDS

Namespace:

External Mapping ID:

External Data Source Type: C

Model

Technical Model Name: ZODATA_DEMO_CDS

Model Version: 1

Creation Information

Package Assignment: \$TMP

Local Object

ICF Node

Standard Mode (selected) None

Set Current Client as Default Client in ICF Node (checked)

OAuth enablement

Enable OAuth for Service (unchecked)

7. A dialog box appears as shown below, click on OK to continue.

Information

Service 'ZODATA_DEMO_CDS' was created and its metadata was loaded successfully.

OK Cancel

8. As a result OData service is activated successfully in the Gateway hub system. Launch the transaction code "/IWFND/GW_CLIENT" to test the service.

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
<?xml version="1.0" encoding="UTF-8"?>
<edmx:Edmx xmlns:edmx="http://schemas.microsoft.com/ado/2007/06/edmx" xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata"
  Version="1.0">
  <edmx:Reference xmlns:edmx="http://docs.oasis-open.org/odata/ns/edmx"
    Uri="http://localhost:8001/sap/opu/odata/INFND/CATALOGSERVICE;v=2/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMM')/Vocabularies(TechnicalName='%2FIWBEP%2FVOC_COMM')/$metadata" />
  <edmx:Include Namespace="com.sap.vocabularies.Common.v1" Alias="Common" />
  </edmx:Reference>
  <edmx:DataService m:DataServiceVersion="2.0">
    <Schema xmlns="http://schemas.microsoft.com/ado/2008/09/edm" Namespace="sap.com.cds_zodata_demo" xml:lang="en" sap:schema-version="1">
      <EntityType Name="ZODATA_DEMOTYPE" sap:label="OData service on ABAP CDS View" sap:content-version="1">
        <Key>
          <PropertyRef Name="AirlineCode" />
        </Key>
        <Property Name="AirlineCode" Type="Edm.String" Nullable="false" MaxLength="3" sap:display-format="UpperCase" sap:label="Airline" />
        <Property Name="AirlineName" Type="Edm.String" MaxLength="20" sap:label="Airline" />
        <Property Name="Currency" Type="Edm.String" MaxLength="5" sap:label="Airline Currency" sap:semantics="currency-code" />
        <Property Name="AirlineURL" Type="Edm.String" MaxLength="255" sap:label="URL" />
      </EntityType>
      <EntityContainer Name="sap.com.cds_zodata_demo_Entities" m:IsDefaultEntityContainer="true" sap:supported-formats="atom json xml" />
      <EntitySet Name="ZODATA_DEMO" EntityType="sap.com.cds_zodata_demo.ZODATA_DEMOTYPE" sap:creatable="false" sap:updatable="false" sap:deletable="false" />
    </Schema>
  </edmx:DataService>
</edmx:Edmx>
```

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is the metadata generated for the OData service created based on the ABAP CDS view. Lets closely look at metadata and identify the different artifacts as part of the OData service.

Entity Type – ZODATA_DEMOTYPE

Entity Set – ZODATA_DEMO

lets see the data by accessing the entity set

http://<server>:8001/sap/opu/odata/sap/ZODATA_DEMO_CDS/ZODATA_DEMO

Example #2 – ABAP CDS View with Parameters

1. Below is the code for ABAP CDS view with parameters

```
1. @AbapCatalog.sqlViewName: 'ZV_ODATA_DEMO_P'
2. @AbapCatalog.compiler.compareFilter: true
3. @AccessControl.authorizationCheck: #NOT_REQUIRED
4. @EndUserText.label: 'OData service on ABAP CDS View'
5.
6. @OData.publish: true
7. define view ZODATA_DEMO_PARAMS
8. with parameters p_carrid :S_CARR_ID
9. as select from scarr
10.
11. {
12. key carrid      as AirlineCode,
13.   carrname      as AirlineName,
14.   currcode      as Currency,
15.   url           as AirlineURL
16. }
17. where carrid = $parameters.p_carrid
```

NOTE: As of now there is no annotation/a way to make the input parameters OPTIONAL in ABAP CDS Views. So you have to pass values for all input parameters in ABAP CDS view while accessing the data from it.

2. Repeat the steps explained above in the Example #1 to create the OData service for this CDS view also.

3. On successfully OData service generation, execute the OData service and look at the metadata generated and following are the 2 entity sets generated

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```

<edmx:DataService m:DataServiceVersion="2.0">
  <Schema xmlns="http://schemas.microsoft.com/ado/2008/09/edm" Namespace="sap.com.cds_zodata_demo_params" xml:lang="en">
    1 <EntityType Name="ZODATA_DEMO_PARAMSType" sap:label="OData service on ABAP CDS View" sap:content-version="1">
      <Key>
        <PropertyRef Name="p_carriid"/>
        <PropertyRef Name="AirlineCode"/>
      </Key>
      <Property Name="p_carriid" Type="Edm.String" Nullable="false" MaxLength="3" sap:label="Airline"/>
      <Property Name="AirlineCode" Type="Edm.String" Nullable="false" MaxLength="3" sap:display-format="UpperCase" sap:label="Airline"/>
      <Property Name="AirlineName" Type="Edm.String" MaxLength="20" sap:label="Airline"/>
      <Property Name="Currency" Type="Edm.String" MaxLength="5" sap:label="Airline Currency" sap:semantics="currency-code"/>
      <Property Name="AirlineURL" Type="Edm.String" MaxLength="255" sap:label="URL"/>
    </EntityType>
    2 <EntityType Name="ZODATA_DEMO_PARAMSParameters" sap:semantics="parameters" sap:content-version="1">
      <Key>
        <PropertyRef Name="p_carriid"/>
      </Key>
      <Property Name="p_carriid" Type="Edm.String" Nullable="false" MaxLength="3" sap:parameter="mandatory" sap:label="Air" sap:filterable="false"/>
      <NavigationProperty Name="Set" Relationship="sap.com.cds_zodata_demo_params.assoc_C853DEABE978CA1E9AD6E72BED83D2BB" ToRole="ToRole_assoc_C853DEABE978CA1E9AD6E72BED83D2BB"/>
    </EntityType>
    <Association Name="assoc_C853DEABE978CA1E9AD6E72BED83D2BB" sap:content-version="1">
      <End Type="sap.com.cds_zodata_demo_params.ZODATA_DEMO_PARAMSParameters" Multiplicity="1" Role="FromRole_assoc_C853D" />
      <End Type="sap.com.cds_zodata_demo_params.ZODATA_DEMO_PARAMSType" Multiplicity="*" Role="ToRole_assoc_C853DEABE978C" />
    </Association>
    <EntityContainer Name="sap.com.cds_zodata_demo_params.Entities" m:IsDefaultEntityContainer="true" sap:supported-formats="application/json,application/xml">
      <EntitySet Name="ZODATA_DEMO_PARAMSSet" EntityType="sap.com.cds_zodata_demo_params.ZODATA_DEMO_PARAMSType" sap:creatable="true" sap:addressable="false" sap:content-version="1"/>
      1 <EntitySet Name="ZODATA_DEMO_PARAMSParameters" EntityType="sap.com.cds_zodata_demo_params.ZODATA_DEMO_PARAMSParameters" sap:creatable="true" sap:addressable="false" sap:content-version="1"/>
    </EntityContainer>
  </Schema>
</edmx:DataService>

```

TA_DEMO_PARAMSSet – Entity set which has 2 key predicates, one is the input parameter and the second one is the key of ABAP CDS View. You need to pass these 2 key predicates while accessing the data

TA_DEMO_PARAMS – Entity set which has 1 key predicate, which is input parameter. You need to pass this key predicate to access the data.

To access the data you can use both entity sets and let's look how we can form the URI for both of them.

If you like to use entity set `ZODATA_DEMO_PARAMSSet` following the URI to be used

`/sap/opu/odata/sap/ZODATA_DEMO_PARAMS_CDS/ZODATA_DEMO_PARAMSSet(p_carriid='AA',AirlineCode='AA')`

```

<entry xmlns="http://www.w3.org/2005/Atom" xmlns:m="http://schemas.microsoft.com/ado/2007/08/dataservices/metadata" xmlns:d="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme">
  <id>http://sap/opu/odata/sap/ZODATA_DEMO_PARAMS_CDS/ZODATA_DEMO_PARAMSSet(p_carriid='AA',AirlineCode='AA')</id>
  <title type="text">ZODATA_DEMO_PARAMSSet(p_carriid='AA',AirlineCode='AA')</title>
  <updated>2017-06-25T14:21:42Z</updated>
  <category term="sap.com.cds_zodata_demo_params.ZODATA_DEMO_PARAMSType" scheme="http://schemas.microsoft.com/ado/2007/08/dataservices/scheme"/>
    <link href="ZODATA_DEMO_PARAMSSet(p_carriid='AA',AirlineCode='AA')" rel="self" title="ZODATA_DEMO_PARAMSType"/>
  <content type="application/xml">
    <m:properties>
      <d:p_carriid>AA</d:p_carriid>
      <d:AirlineCode>AA</d:AirlineCode>
      <d:AirlineName>American Airlines</d:AirlineName>
      <d:Currency>USD</d:Currency>
      <d:AirlineURL>http://www.aa.com</d:AirlineURL>
    </m:properties>
  </content>
</entry>

```

If you like to use entity set `ZODATA_DEMO_PARAMS` following the URI to be used

`/sap/opu/odata/sap/ZODATA_DEMO_PARAMS_CDS/ZODATA_DEMO_PARAMS(p_carriid='AA')/Set`



I liked it, please share it! Thanks!

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